

## **Week 13: Where the Gap Comes From**

Hankel\_AD and Hankel\_CDC\_AD, 23–30 September

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The gap to SOTA was **one dataset family**.

Locality fixes it – but not yet without test data.

Local tangent geometry (CDC) did not beat plain distance.

## Corrections to earlier claims

- “max\_d cap ties unclamped” (Week 12) – **false**
- Golyandina & Zhigljavsky (2011) – **does not exist**
- “1.8 points behind SOTA” – **misleading framing**
- Ensemble’s Eva gain – **depended on test-period scaling**

**Hankel\_AD**

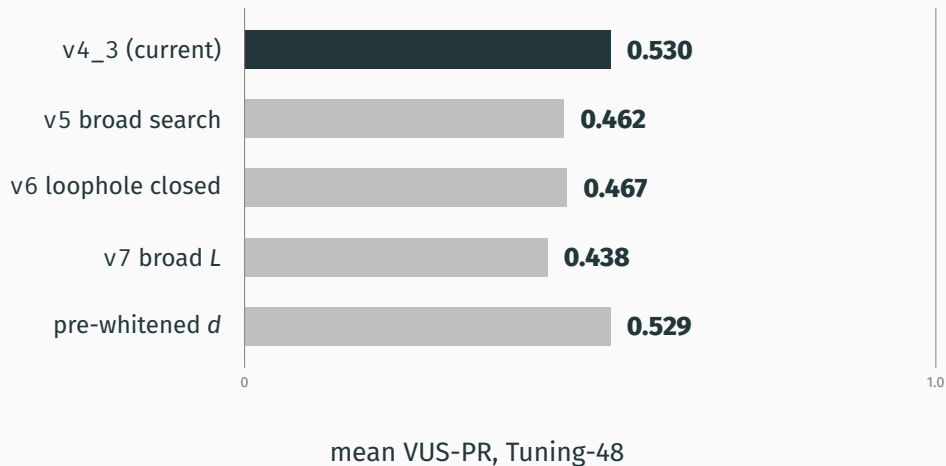
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## How much is left on the table?



mean VUS-PR, Eva · oracle picks  $d \approx 50$ , we pick  $d \approx 11$

## Four redesigns of $(L, d)$ , four failures



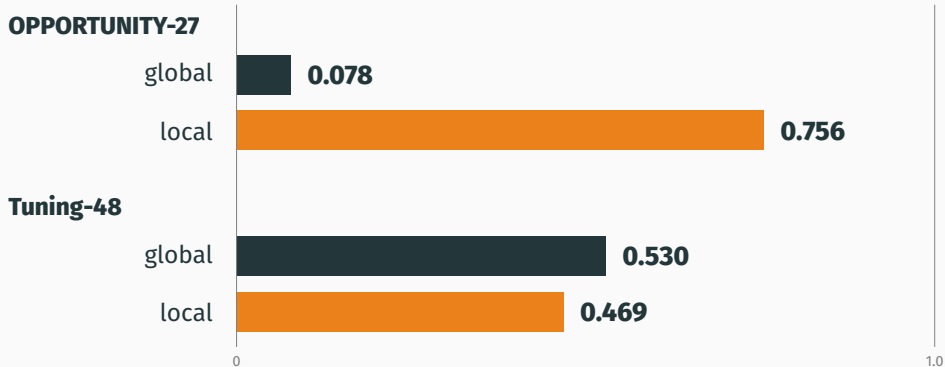
# The gap is one family

OPPORTUNITY (HumanActivity) – 27 of 350 datasets



**27 / 27 lose** · **excluding them:  $-0.020 \rightarrow +0.034$**

## Locality fixes it – a straight swap breaks the rest



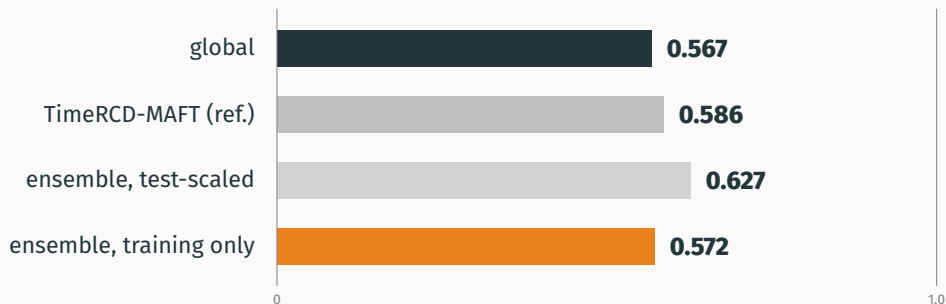
## Ensemble: take the more confident scorer, per window

$$z_s(w) = \frac{\text{score}_s(w) - \text{median}(\text{score}_s[\text{train}])}{1.4826 \cdot \text{MAD}(\text{score}_s[\cdot])}, \quad s \in \{\text{global}, \text{local}\}$$

$$\text{ensemble}(w) = \max(z_{\text{global}}(w), z_{\text{local}}(w))$$

MAD over train  $\cup$  test (first version)   or   training only (corrected)

## Official Eva split ( $n = 350$ ): the gain needs test-period scaling



mean VUS-PR · training-only vs. global:  $+0.005$ , Wilcoxon  $p = 0.18$

## Where the gain goes

| ensemble – global (mean VUS-PR) | all 350                     | excl. OPPORTUNITY (323) |
|---------------------------------|-----------------------------|-------------------------|
| test-scaled                     | +0.060 $p=5 \times 10^{-5}$ | +0.010 $p=0.051$        |
| <b>training only</b>            | + <b>0.005</b> $p=0.18$     | – <b>0.046</b> $p=0.46$ |

OPPORTUNITY-27, training only: 0.080  $\rightarrow$  0.685 – the fix itself needs no test data

- Not blind: OPPORTUNITY is inside the Eva split
- Test-scaled result is transductive; training-only gain is n.s.
- SOTA figure is a reference, not a rerun
- 4 OPPORTUNITY datasets fail under every local method

**Hankel\_CDC\_AD**

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## Synthetic ground truth: tangents exact, curvature wrong

|                    | tangent-plane error | curvature (true = 1) |
|--------------------|---------------------|----------------------|
| sphere, $n = 300$  | $\leq 0.0015$       | 4.2                  |
| sphere, $n = 2000$ | $\leq 0.0002$       | 24.0                 |
| sphere, $n = 5000$ | $\leq 0.0002$       | 59.1                 |

tangent score: AUC 1.0 on synthetic off-manifold anomalies

## Real data: CDC does not beat plain distance



Tuning: 14 wins – 14 losses – 2 ties ( $p = 0.927$ )

## Fixed

- GPU port:  $8.5\times$  faster
- three memory bottlenecks
- scoring new points

## Open

- results differ between sessions
- caps move VUS-PR by up to 0.33
- $n_{\text{train}}$  capped at 4000

- Combine local + global without test data, without harming the rest
- Then a blind re-validation
- Cap-free CDC reruns before quoting its aggregates
- Finish the citation clean-up